

The Multilinear Extension

Past lectures: convex closure - coincides w/ Lovász
concave closure Ext. when f is submodular.
 $\downarrow$
in general hard to evaluate

Given: $f: \{0,1\}^E \rightarrow \mathbb{R}$ How to extend to $[0,1]^E$?

Idea: interpret $x \in [0,1]^E$ as a product dist'n on E elts.

$F(x) = \mathbb{E}[f(\hat{x})]$, $\hat{x}$ drawn according to x

i.e., $P(\hat{x}_i = 1) = x_i$

Def'n: Multi-linear Extension

Given $f: \{0,1\}^E \rightarrow \mathbb{R}$, define $F(x) = \sum_{S \subseteq E} f(S) \cdot \prod_{i \in S} x_i \cdot \prod_{j \notin S} (1-x_j)$

Note: $\frac{\partial^2 F}{\partial x_i^2} = 0$

Properties for Monotone/Submodular

Lemma: (i) If f is monotone, then $\frac{\partial F}{\partial x_i} \geq 0$ ✓
 $\forall i, \forall x \in [0,1]^E$.

(ii) If f is submodular, then

$$\frac{\partial^2 F}{\partial x_i \partial x_j} \leq 0 \quad \forall i, j, \quad \forall x \in [0,1]^E.$$

pf: (i) $F(x) = \sum_{S \subseteq E} f(S) \cdot \prod_{i \in S} x_i \cdot \prod_{j \notin S} (1-x_j)$

$$= \left[\sum_{\substack{S \subseteq E \\ k \notin S}} f(S) \cdot \prod_{i \in S} x_i \cdot \prod_{j \in S} (1-x_j) \right] x_k + \left[\sum_{\substack{S \subseteq E \\ k \in S}} f(S) \cdot \prod_{i \in S} x_i \cdot \prod_{\substack{j \in S \\ j \neq k}} (1-x_j) \right] (1-x_k)$$

$$\begin{aligned} \frac{\partial F}{\partial x_k} &= F(x_{-k}, x_k=1) - F(x_{-k}, x_k=0) \\ &= \mathbb{E}[f(R \cup k) - f(R \setminus k)] \geq 0 \end{aligned}$$

$R \subseteq E$ random s.t.
 $i \in R$ w.p. x_i .

Proof of Lemma — continued —

The proof of (ii) is similar.

$\frac{\partial F}{\partial x_i}$ are also multilinear. And by similar

calculation:

$$\begin{aligned}\frac{\partial^2 F}{\partial x_i \partial x_j} &= \frac{\partial F}{\partial x_j} \Big|_{(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_E)} - \frac{\partial F}{\partial x_j} \Big|_{(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_E)} \\ &= \mathbb{E} \left[\underbrace{f(R_{U\{i,j\}})} - \underbrace{f(R_{U\{i\} \cup j})} \right] - \mathbb{E} \left[\underbrace{f(R_{\{i\} \cup j})} - \underbrace{f(R_{\{i\} \cup j})} \right] \\ &= \mathbb{E} \left[\underbrace{f_{R_{U\{i\}}}(j)} - \underbrace{f_{R_{\{i\}}}(j)} \right] \leq 0\end{aligned}$$

Convexity/Concavity Properties

Corollary:

(i) f monotone $\Rightarrow F$ non-decreasing along any $d \geq 0$

(ii) f submodular $\Rightarrow F$ is concave along any $d \geq 0$

(iii) f submodular $\Rightarrow F$ is convex along $d = e_i - e_j$.

pf: $\phi(t) = F(x + td)$

$$\phi'(t) = \sum d_i \cdot \frac{\partial F}{\partial x_i} \Big|_{x+td}$$

so (i) follows since $\frac{\partial F}{\partial x_i} \geq 0$, $d_i \geq 0$.

$$\phi''(t) = \sum_{i,j} d_i d_j \frac{\partial^2 F}{\partial x_i \partial x_j} \Big|_{x+td}$$

so (ii) follows since $\frac{\partial^2 F}{\partial x_i \partial x_j} \leq 0$, $d_i \geq 0$.

If $d = e_i - e_j$

$$\phi''(t) = \frac{\partial^2 F}{\partial x_i^2} \Big|_{x+td} - 2 \frac{\partial^2 F}{\partial x_i \partial x_j} \Big|_{x+td} + \frac{\partial^2 F}{\partial x_j^2} \Big|_{x+td} \geq 0$$

Hence (iii) follows.

Approximate Evaluation

we cannot evaluate $F(x) = \sum_{S \in E} f(S) \prod_{i \in S} x_i \cdot \prod_{j \notin S} (1-x_j)$

efficiently (poly # of evaluations of $f(S)$).

Since: $F(x) = \mathbb{E}[f(\hat{x})]$ we can evaluate this expectation by taking iid samples and averaging.

Indeed: $R_i \in E$ sampled iid according to x .

$$M = \max |f(S)|$$

$f(R)$ r.v. in $[-M, M]$

$$Z_i = \frac{1}{M} f(R_i)$$

$$\Rightarrow \mathbb{P}\left(\left|\sum_{i=1}^t Z_i - \frac{t}{M} F(x)\right| \geq t \cdot \epsilon\right) \leq \exp(-t \epsilon^2/4)$$

Lemma: If we sample $R_1, \dots, R_t$ iid $\sim x$, then

$$\left|\frac{1}{t} \sum f(R_i) - F(x)\right| \leq \epsilon \cdot \max |f(S)|, \quad \text{w.p.} \geq 1 - \exp\{-t \epsilon^2/4\}$$